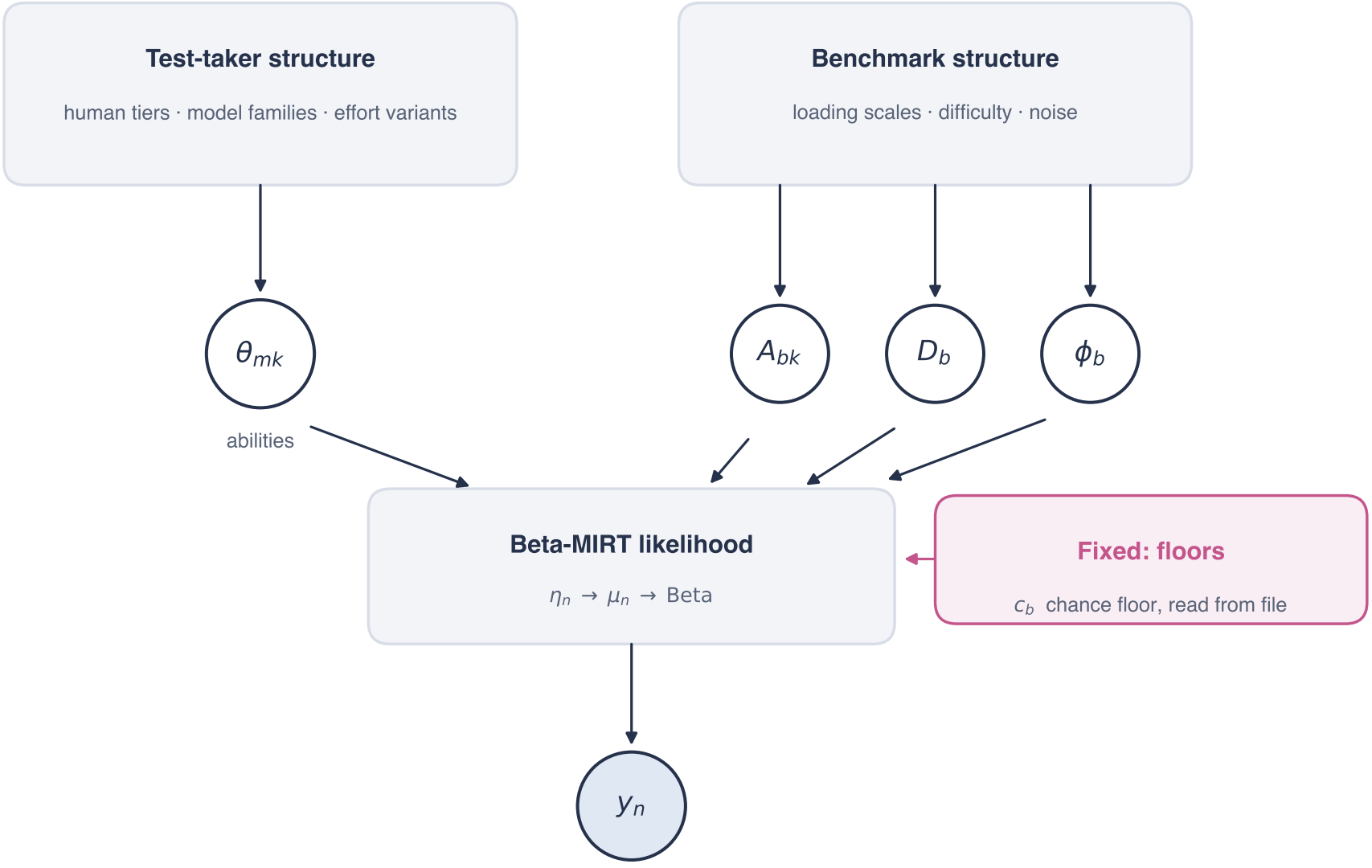
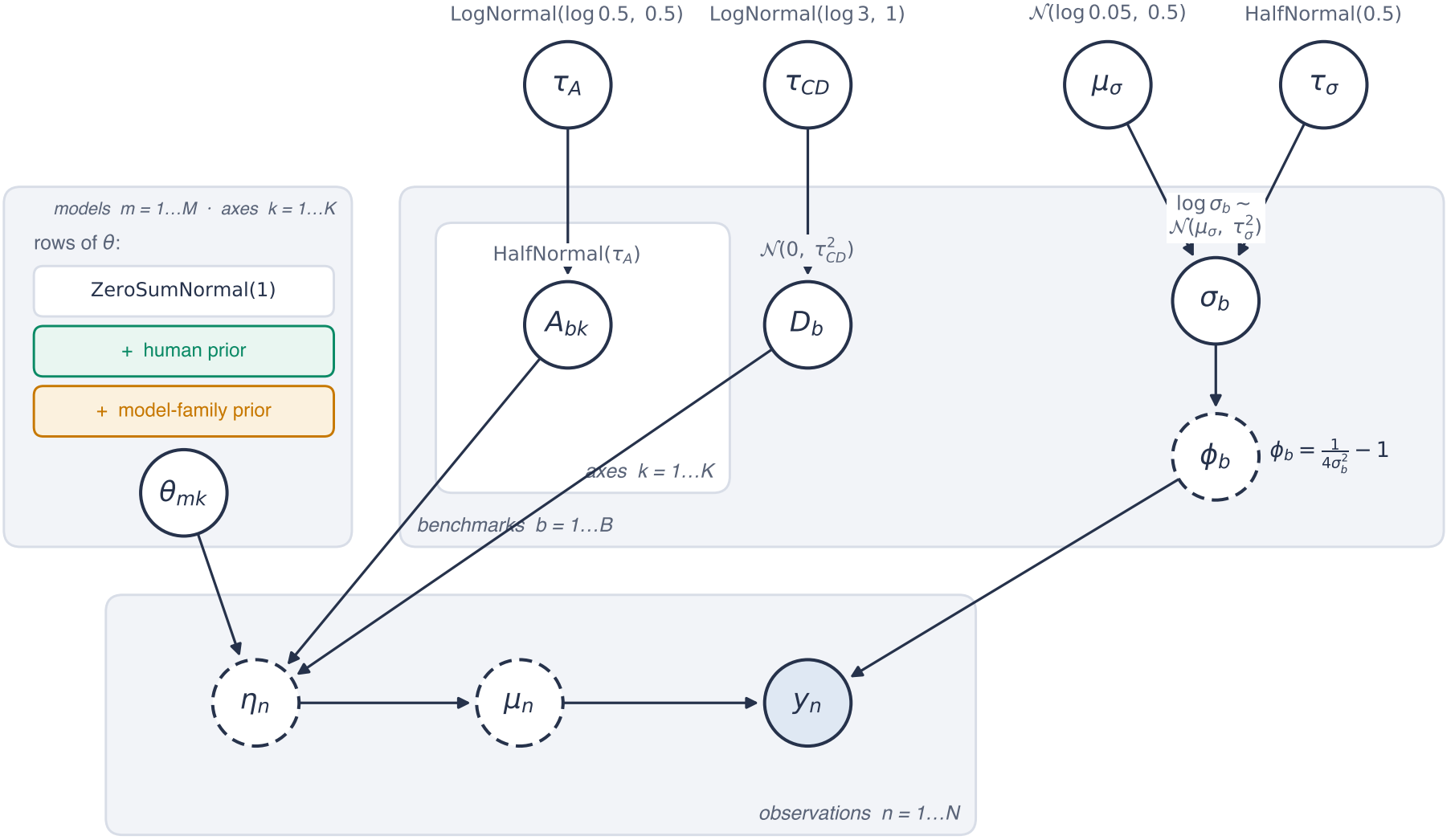
Overview


Test-taker structure
human tiers · model families · effort variants
Benchmark structure
loading scales · difficulty · noise
θ_{mk}
abilities
A_{bk}
D_b
ϕ_b
Beta-MIRT likelihood
$\eta_n \to \mu_n \to$ Beta
Fixed: floors
c_b chance floor, read from file
y_n

Beta-MIRT model (K = 4)



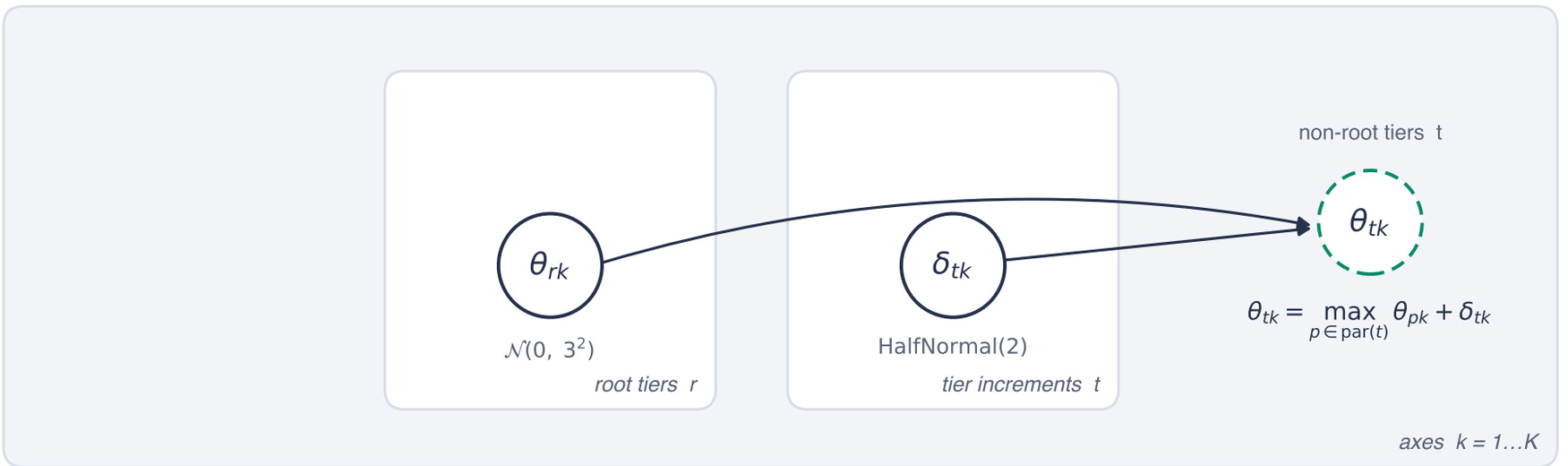
$$\eta_n = \sum_k A_{b_n k}\,\theta_{m_n k} - D_{b_n}$$

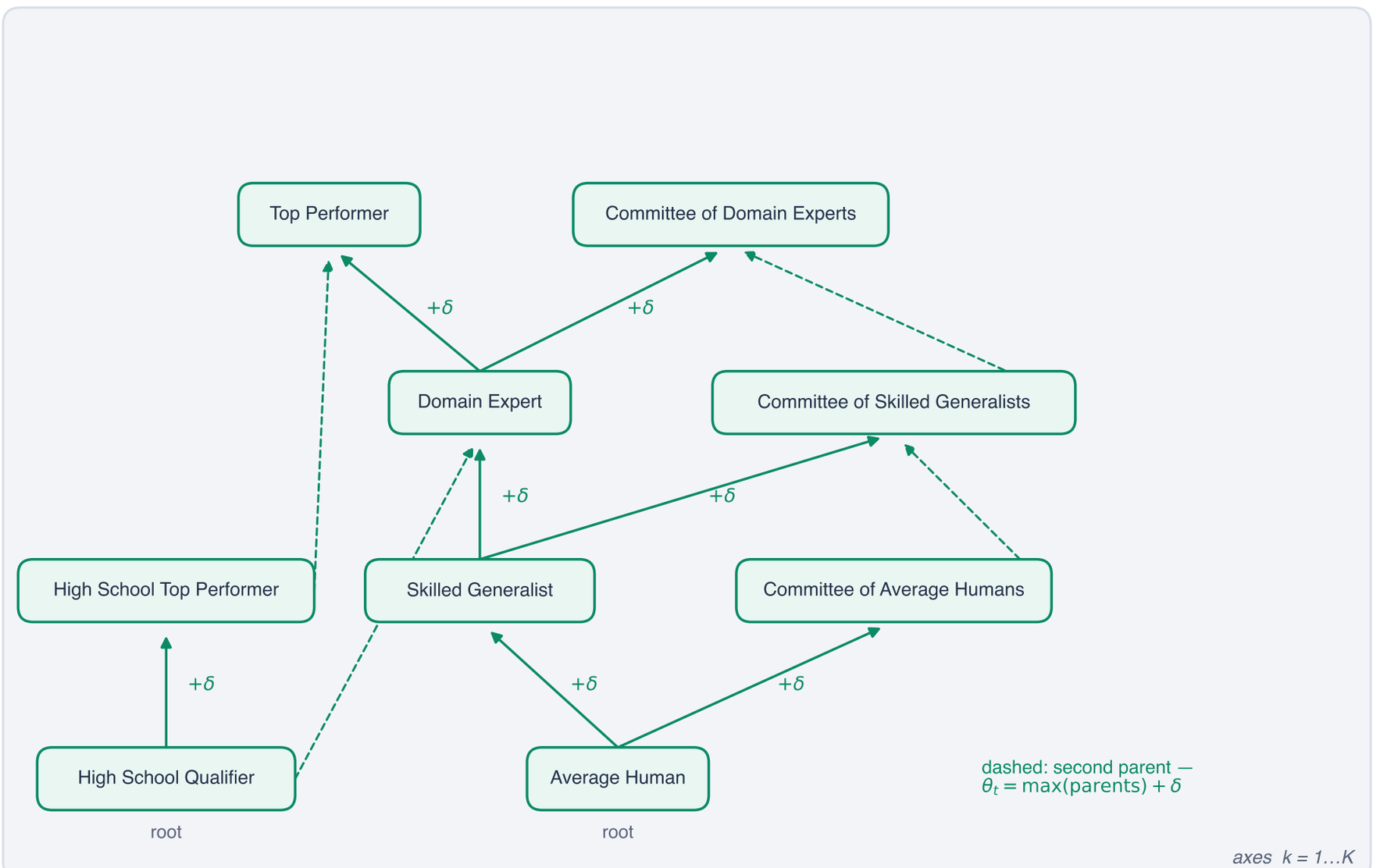
$$\mu_n = c_{b_n} + (1 - c_{b_n})\,\sigma(\eta_n) \qquad c_b\text{: chance floor, fixed}$$

$$y_n \sim \mathrm{Beta}(\mu_n \phi_{b_n},\ (1-\mu_n)\phi_{b_n})$$

stochastic · deterministic · observed · fixed constant

Human prior


non-root tiers t
θ_{rk}
$\mathcal{N}(0,\ 3^2)$
root tiers r
δ_{tk}
HalfNormal(2)
tier increments t
θ_{tk}
$\theta_{tk} = \max_{p \in \mathrm{par}(t)} \theta_{pk} + \delta_{tk}$
axes k = 1…K


Top Performer
Committee of Domain Experts
+δ
+δ
Domain Expert
Committee of Skilled Generalists
+δ
+δ
High School Top Performer
Skilled Generalist
Committee of Average Humans
+δ
+δ
+δ
High School Qualifier
Average Human
root
root
dashed: second parent —
$\theta_t = \max(\text{parents}) + \delta$
axes k = 1…K

Model-family prior · time-based steps

